



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

SpaceX Falcon 9 First
Stage Landing
Prediction
June 2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Data was collected via SpaceX REST API and web scraping Wikipedia. After wrangling, EDA was performed using SQL, Matplotlib/Seaborn visualizations, and Folium interactive maps. A Plotly Dash dashboard was built. Four ML classifiers (Logistic Regression, SVM, Decision Tree, KNN) were trained and tuned using GridSearchCV with 10-fold cross-validation.
- Decision Tree achieved the best test accuracy of 94.4%. KSC LC-39A had the highest success rate. Higher flight numbers correlated with more successes. Payloads in the 2,000-5,500 kg range showed strong success. The model can reliably predict first-stage landing success, enabling cost estimation for Falcon 9 launches (~\$62M vs ~\$165M without reuse).

Introduction

- SpaceX advertises Falcon 9 rocket launches at \$62M, far less than other providers charging upwards of \$165M. The key cost reduction comes from reusing the first stage. This project uses publicly available SpaceX launch data to predict whether the first stage will successfully land and be reused, enabling cost estimation for competing companies bidding against SpaceX.
- Key questions: (1) What factors determine successful first-stage landing? (2) Which launch site has the highest success rate? (3) How does payload mass affect landing outcome? (4) Which ML model best predicts landing success? (5) Has success rate improved over time?

Section 1

Methodology

Methodology

Executive Summary

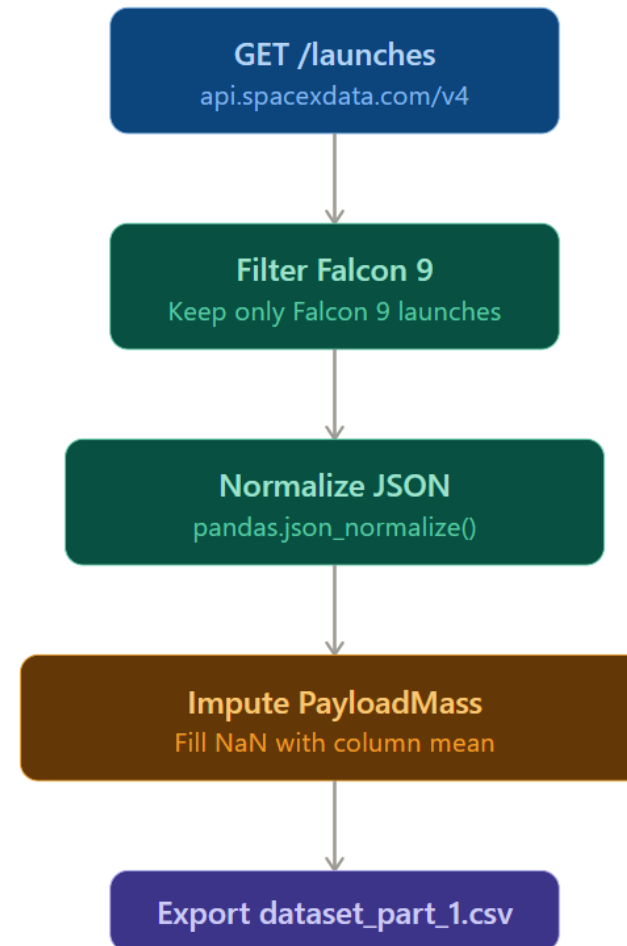
- Data collection methodology:
 - Describe how data was collected
- Perform data wrangling
 - Binary landing class labels created; PayloadMass NaN imputed with mean
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

Data Collection

- Two data collection methods were used: (1) SpaceX REST API - queried launch data including rocket info, payload, launch site, and landing outcomes. (2) Web Scraping - extracted historical Falcon 9 launch records from Wikipedia using BeautifulSoup. Combined dataset contains 90 launch records with 18 features including flight number, payload mass, orbit, launch site, booster version, and landing outcome (Class: 1=success, 0=failure).
- Process: SpaceX API → Parse JSON → Filter Falcon 9 launches → Fill missing PayloadMass with mean (NaN %) → Export CSV. Wikipedia → BeautifulSoup HTML parse → Extract launch table → Clean column headers → Export CSV. Total records: 90 Falcon 9 launches (2010-2020).

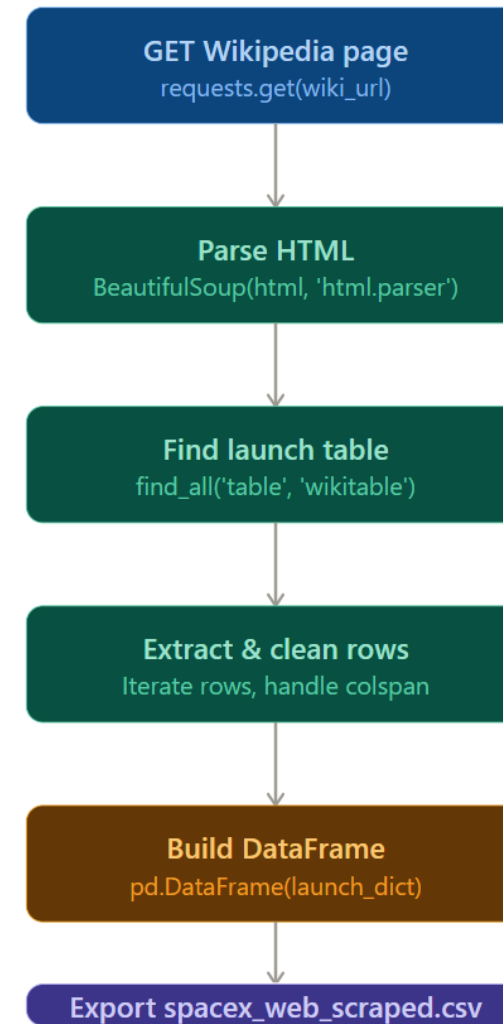
Data Collection – SpaceX API

- Used requests library to call `api.spacexdata.com/v4/rockets` and `/launches`. Extracted: rocket name, payload mass, launch site, flight number, orbit, booster version, reuse flags. Normalized nested JSON into flat DataFrame with `pandas.json_normalize()`.
- https://github.com/lmuazkhanl/IBM-Capstone-Project/blob/main/1__jupyter-labs-spacex-data-collection-api.ipynb



Data Collection - Scraping

- Used BeautifulSoup4 + requests to scrape Falcon 9 launch history from Wikipedia. Steps: (1) GET Wikipedia page → parse HTML → find launch table → extract rows → clean column headers → parse dates/outcomes → export CSV.
- https://github.com/lmuazkhan/IBM-Capstone-Project/blob/main/2__jupyter-labs-web scraping.ipynb



Data Wrangling

- Data wrangling steps: (1) Loaded dataset_part_1.csv (90 records, 18 columns). (2) Checked missing values: PayloadMass had NaN values, imputed with column mean. (3) Converted outcome strings to binary Class labels: True ASDS/RTLS/Ocean = 1 (success), else = 0. (4) Verified landing outcome distribution: True ASDS=41, True RTLS=14, True Ocean=5, failures=various.
- Outcome mapping: bad_outcomes = {None None, False ASDS, False RTLS, False Ocean, None ASDS}. Class=0 if outcome in bad_outcomes, else Class=1. Final label distribution: 64 successes (Class=1), 26 failures (Class=0).
- Launch sites: CCAFS SLC-40 (55 launches), KSC LC-39A (22), VAFB SLC-4E (13). Key orbits: ISS (21), VLEO (14), PO (9), LEO (7), SSO (5).
- https://github.com/ImuazkhanI/IBM-Capstone-Project/blob/main/3__jupyter-labs-spacex-Data%20wrangling.ipynb

EDA with Data Visualization

- **Launch Site Success Over Time (Scatter Plot):** Correlates *FlightNumber* with *LaunchSite*, revealing that success rates improved as the flight program progressed.
- **Payload Capability by Launch Site (Scatter Plot):** Maps *PayloadMass* against *LaunchSite* to pinpoint the specific payload weight ranges handled by each site.
- **Success Rate by Orbit Type (Bar Chart):** Highlights orbital performance, showing a perfect 100% success rate for ES-L1, SSO, VLEO, and GEO orbits.
- **Orbital Trends Over Time (Scatter Plot):** Compares *FlightNumber* against *Orbit* to track how mission targets shifted and evolved chronologically.
- **Payload Capacity vs. Orbit (Scatter Plot):** Analyzes *PayloadMass* across different orbits, proving successful outcomes for VLEO missions at heavier masses.
- **Annual Success Trajectory (Line Chart):** Illustrates the yearly average success rate, highlighting a distinct and steady upward trend from 2013 through 2020.
- https://github.com/lmuazkhan/IBM-Capstone-Project/blob/main/5__jupyter-labs-edadataviz.ipynb

EDA with SQL

- SQL queries on SPACEXTBL (SQLite): (1) SELECT DISTINCT LAUNCH_SITE → 4 unique sites. (2) WHERE LAUNCH_SITE LIKE CCA% LIMIT 5 → 5 CCA records. (3) SUM(PAYLOAD) WHERE CUSTOMER=NASA(CRS) → 45,596 kg. (4) AVG(PAYLOAD) WHERE BOOSTER=F9 v1.1 → 2,534.67 kg. (5) MIN(DATE) WHERE LANDING=Success(ground pad) → 2015-12-22. (6) Boosters with drone ship success & 4000<payload<6000 → 4 boosters. (7) Mission outcomes count: Success=99, Failure=1. (8) Max payload boosters → all F9 B5 variants. (9) 2015 drone ship failures → 2 records. (10) Landing outcome ranks 2010-2017.
- Tool: ipython-sql with SQLite. Key finding: 99 successful missions out of 101 total. Total NASA CRS payload carried: 45,596 kg. First successful ground pad landing: December 22, 2015.
- https://github.com/lmuazkhanl/IBM-Capstone-Project/blob/main/4__jupyter-labs-eda-sql-coursera_sqlite.ipynb

Build an Interactive Map with Folium

- Folium map objects created: (1) CircleMarkers for all 4 launch sites (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E) with popups showing site name. (2) Color-coded markers for each launch outcome (green=success, red=failure). (3) MousePosition plugin for coordinate display. (4) Distance markers (PolyLines) from KSC LC-39A to nearest railway, highway, coastline, and city.
- Markers show geographic distribution of launch sites. Color-coded outcomes reveal that KSC LC-39A has the highest success density. Distance lines show proximity to infrastructure — launch sites are near coastlines for safety but close to transport for logistics.
- Key finding: All 4 launch sites are on the US coastline. KSC LC-39A (Florida) shows the highest success rate visually. Launch sites are strategically placed near highways and railways but safely away from dense urban areas.
- https://github.com/lmuazkhanl/IBM-Capstone-Project/blob/main/6__jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

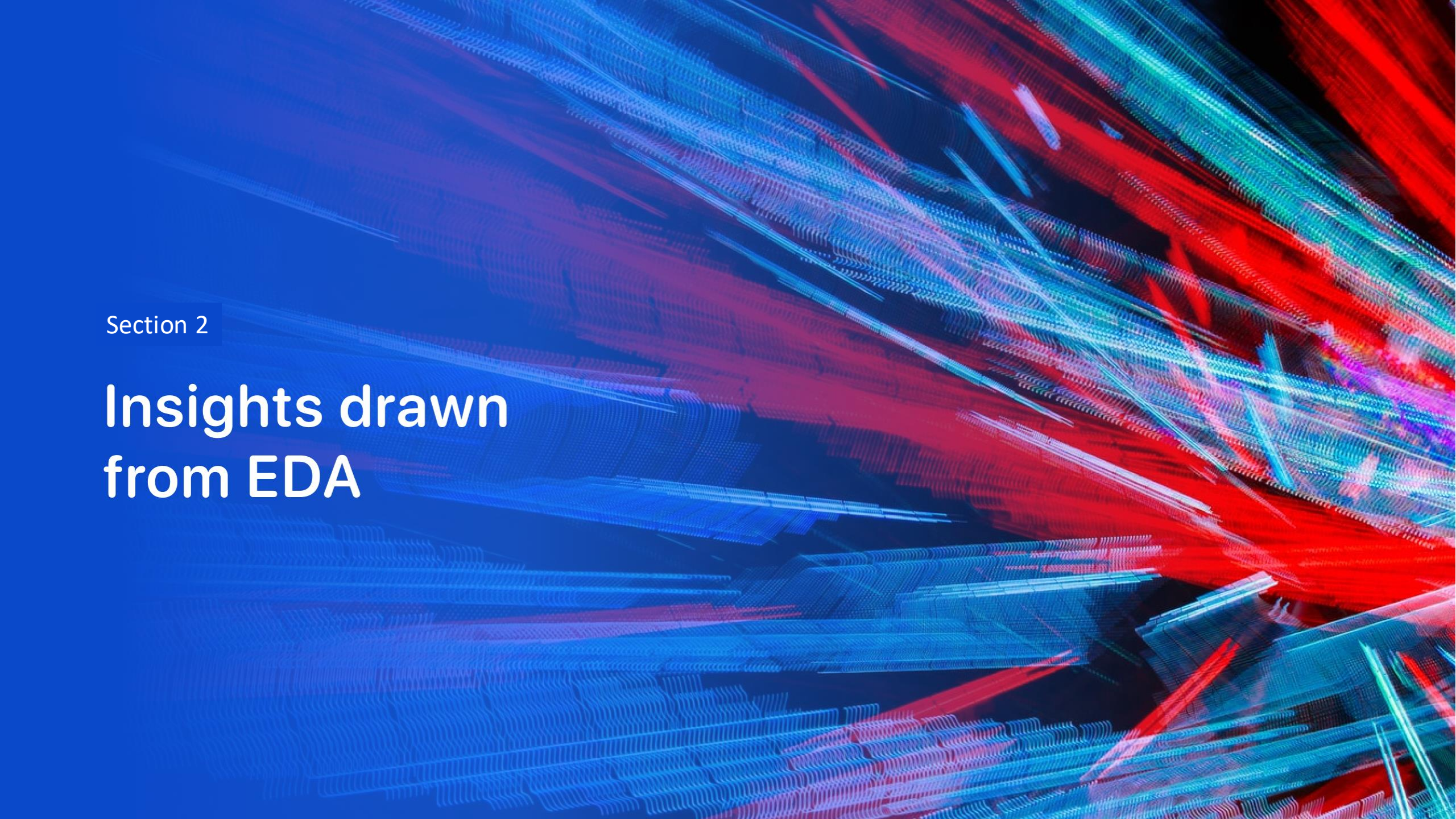
- Plotly Dash dashboard with 4 interactive components: (1) Dropdown: select All Sites or specific launch site. (2) Pie chart: shows total successful launches by site (ALL) or success vs failure ratio for selected site. (3) RangeSlider: filter payload from 0 to 10,000 kg. (4) Scatter plot: Payload Mass vs Launch Outcome colored by Booster Version Category, filtered by site and payload range.
- Pie chart reveals KSC LC-39A has the highest individual success ratio (~76.9%). Scatter plot with payload slider shows FT boosters succeed at mid-range payloads (2,000-6,000 kg). B5 variants handle heavier payloads successfully. Interactive filtering allows stakeholders to explore specific scenarios.
- https://github.com/ImuazkhanI/IBM-Capstone-Project/blob/main/7__spacex-dash-app.py

Predictive Analysis (Classification)

- Pipeline: (1) Standardize features with StandardScaler. (2) Split 80/20 train/test (stratified). (3) Train 4 models: Logistic Regression, SVM, Decision Tree, KNN. (4) Tune each with GridSearchCV (10-fold CV). (5) Evaluate on test set.
- Flow: Raw Data → StandardScaler → Train/Test Split (80/20) → GridSearchCV (4 models, 10-fold CV) → Best Parameters → Test Evaluation → Decision Tree wins with 94.4% accuracy
- Decision Tree outperformed all models. Confusion matrix showed strong performance predicting both landing success and failure. GridSearchCV with 10-fold cross-validation prevented overfitting and ensured robust hyperparameter selection.

Results

- EDA: Flight # vs Launch Site, Payload vs Orbit, Yearly Success Trend
- Folium map with 4 launch sites + color-coded outcomes; Plotly Dash with pie chart & scatter
- Decision Tree: 94.4% accuracy; beats LR, SVM, KNN (all ~83.3%)

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is high-tech and digital.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

- FINDING: Later flights show higher success. CCAFS SLC-40 has most launches. After flight 60, success dominates all sites. KSC LC-39A launched later but with immediate high success.
- X-axis: Flight Number (1-90), Y-axis: Launch Site (4 sites), Color: Green=Success, Red=Failure. Pattern shows SpaceX improved reliability significantly over time at all sites.

Payload vs. Launch Site

- FINDING: CCAFS handles widest payload range. KSC handles heaviest loads (4,000-16,000 kg). VAFB handles lighter polar payloads. Mid-range payloads 2,000-6,000 kg show best success rates.
- X-axis: Payload Mass (kg), Y-axis: Launch Site, Color: Green=Success, Red=Failure. Mid-range payloads (2,000-6,000 kg) show best success rates. Very light or very heavy payloads had lower success in early missions.

Success Rate vs. Orbit Type

- Show a FINDING: ES-L1, GEO, HEO, SSO, VLEO, SO all have 100% success. GTO ~52% due to high energy demands. ISS ~76%, LEO ~80%, PO ~89%, MEO ~100%.
- Bar chart: X-axis = Orbit type, Y-axis = Success rate (0-1.0). High-energy GTO orbit shows lower success due to early mission challenges. Polar and sun-synchronous orbits (SSO, PO) are very reliable. This guides future mission planning.

Flight Number vs. Orbit Type

- Show a FINDING: GTO missions span all flight numbers. ISS missions clustered in flights 1-60. VLEO (Starlink) missions in later flights (40+). After flight 40, success dominates all orbit types.
- X-axis: Flight Number, Y-axis: Orbit Type, Color: Green=Success, Red=Failure. Early failures in ISS and LEO orbits. GTO failures mostly in flights 1-30. After flight 40, success became dominant across all orbit types.

Payload vs. Orbit Type

- Show a FINDING: GTO payloads ~4,000-6,000 kg. VLEO (Starlink) heaviest up to 15,600 kg with high success. ISS range 0-12,000 kg. Failures concentrated at lower masses in early GTO missions.
- X-axis: Payload Mass (kg), Y-axis: Orbit Type, Color: Green=Success, Red=Failure. Heavier VLEO/Starlink payloads show high success with newer B5 booster. Failures concentrated at lower masses in early GTO missions.

Launch Success Yearly Trend

- Show a FINDING: Clear upward trend 2013-2020. 2010-2012: 0% landing success. 2015-2016: ~20-40% as recovery tests began. 2018-2020: 90%+ success. SpaceX mastered reusability by 2020.
- X-axis: Year, Y-axis: Average Success Rate (0-1.0). Rapid improvement after 2015 reflects engineering improvements and booster reuse. By 2020, Falcon 9 landing had become highly reliable, validating SpaceX's reusability strategy.

All Launch Site Names

- SQL: `SELECT DISTINCT LAUNCH_SITE FROM SPACEXTBL`
- Result: 4 unique launch sites found: CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40. These are all located on the US coastline — two in Florida (CCAFS) and one each in California (VAFB) and Kennedy Space Center (KSC).

Launch Site Names Begin with 'CCA'

- SQL: `SELECT * FROM SPACEXTBL WHERE LAUNCH_SITE LIKE 'CCA%' LIMIT 5`
- Result: 5 records from CCAFS LC-40. First launch: 2010-06-04 (F9 v1.0 B0003, Dragon Qualification, 0 kg, LEO). Second: 2010-12-08 (F9 v1.0 B0004, Dragon demo C1, 0 kg, LEO/ISS). Pattern: early Falcon 9 flights from CCAFS LC-40.

Total Payload Mass

- SQL: `SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE CUSTOMER = 'NASA (CRS)'`
- Result: 45,596 kg total payload carried for NASA CRS missions. This represents SpaceX's Commercial Resupply Service cargo delivered to the International Space Station. Multiple Dragon spacecraft missions contribute to this total.

Average Payload Mass by F9 v1.1

- SQL: `SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE BOOSTER_VERSION LIKE 'F9 v1.1%'`
- Result: 2,534.67 kg average payload for F9 v1.1 booster. F9 v1.1 was an upgraded version with more powerful Merlin 1D engines, enabling heavier payloads than the original v1.0 (which typically carried 0 kg on test flights).

First Successful Ground Landing Date

- SQL: `SELECT MIN(DATE) FROM SPACEXTBL WHERE LANDING_OUTCOME = 'Success (ground pad)'`
- Result: 2015-12-22 — December 22, 2015. This historic date marks the first successful Falcon 9 first-stage return and vertical landing at Cape Canaveral Air Force Station Landing Zone 1 (LZ-1). This was a major milestone in SpaceX's reusability program.

Successful Drone Ship Landing with Payload between 4000 and 6000

- SQL: SELECT BOOSTER_VERSION FROM SPACEXTBL WHERE LANDING_OUTCOME='Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000
- Result: 4 boosters — F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2. All are Full Thrust (FT) variants. Drone ship landings (ASDS) are used for heavier payloads where the booster lacks fuel to return to launch site. This payload range is typical for commercial GTO missions.

Total Number of Successful and Failure Mission Outcomes

- SQL: `SELECT Mission_Outcome, COUNT(*) FROM SPACEXTBL GROUP BY TRIM(Mission_Outcome)`
- Result: Success = 99 missions, Failure (in flight) = 1 mission, Success (payload status unclear) = 1 mission. Total: 101 records. The single in-flight failure occurred on June 28, 2015 (CRS-7). SpaceX's overall mission success rate is approximately 98%.

Boosters Carried Maximum Payload

- SQL: `SELECT BOOSTER_VERSION FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL)`
- Result: 12 boosters, all F9 B5 variants (B1048.4, B1049.4, B1051.3, B1056.4, B1048.5, B1051.4, B1049.5, B1060.2, B1058.3, B1051.6, B1060.3, B1049.7). These are Starlink missions with maximum payload mass. The F9 B5 (Block 5) is SpaceX's final, most capable Falcon 9 variant.

2015 Launch Records

- List the failed WHERE LANDING_OUTCOME='Failure (drone ship)' AND in drone ship, their booster versions, and launch site names for in year 2015
- Result: 2 failures in 2015. (1) January: F9 v1.1 B1012, CCAFS LC-40 — attempted ASDS landing, missed drone ship. (2) April: F9 v1.1 B1015, CCAFS LC-40 — another ASDS miss. Both failures were early drone ship attempts; SpaceX refined landing algorithms after these events.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- SQL: SELECT LANDING_OUTCOME, COUNT(*) AS COUNT FROM SPACEXTBL WHERE DATE BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY LANDING_OUTCOME ORDER BY COUNT DESC
- Result (ranked): No attempt=10, Success (drone ship)=5, Failure (drone ship)=5, Success (ground pad)=3, Controlled (ocean)=3, Uncontrolled (ocean)=2, Failure (parachute)=2, Precluded (drone ship)=1. Early period shows equal drone ship successes and failures, reflecting the learning curve from 2015-2017.

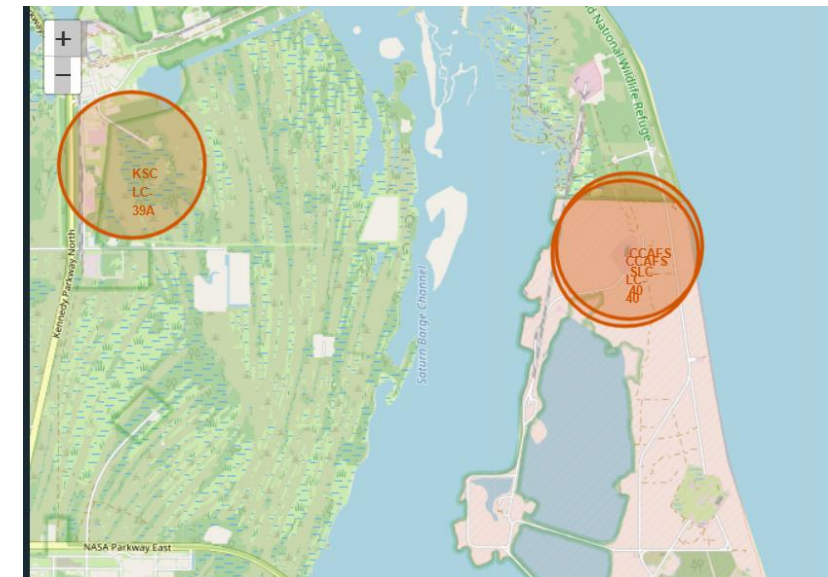
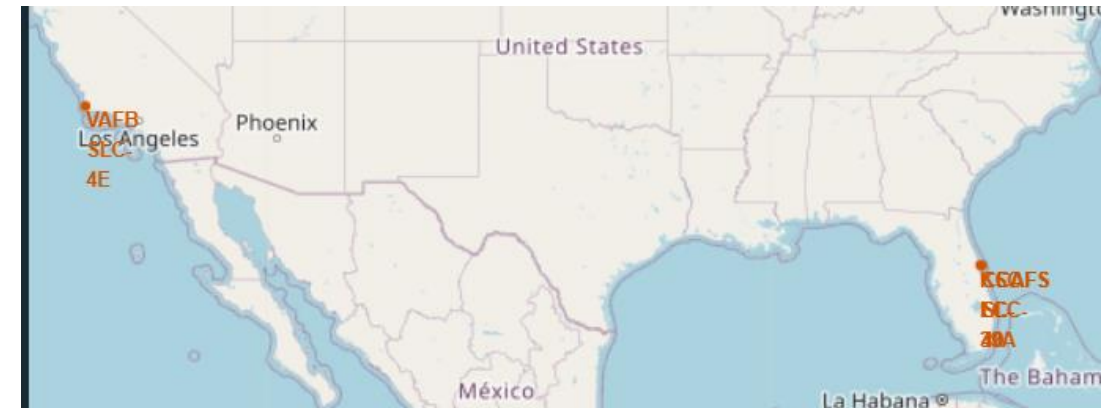
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible as a curved line separating the dark surface from the deep blue of space.

Section 3

Launch Sites Proximities Analysis

Global map showing CircleMarkers at CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E]

- Launch Sites on Global Map — Folium CircleMarkers
- All 4 SpaceX launch sites plotted with CircleMarkers. Sites: CCAFS LC-40 (Florida), CCAFS SLC-40 (Florida), KSC LC-39A (Kennedy Space Center), VAFB SLC-4E (California). Each marker displays popup with site name and coordinates.
- KEY FINDING: All US launch sites are on the coastline for safety (rocket trajectory over ocean). Two Florida sites (CCAFS) and one California site (VAFB) cover Eastern and Western launch trajectories. KSC supports heavier payloads to lower inclination orbits.

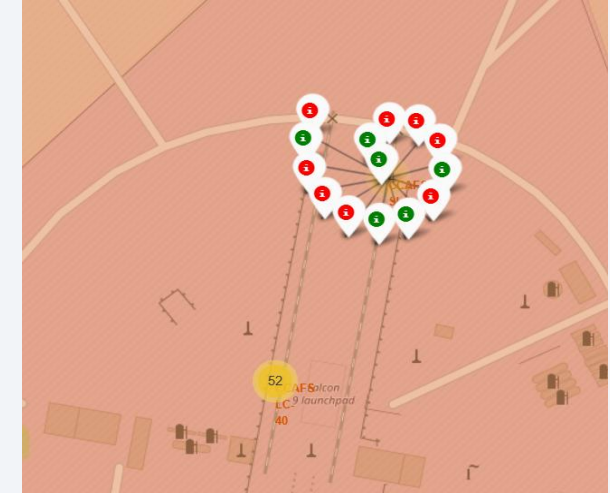


Color-Coded Launch Outcomes on Map

- Each launch plotted with color-coded markers: Green=Success (Class 1), Red=Failure (Class 0). Multiple markers at each site.
- KEY FINDING: KSC LC-39A shows the highest density of green (successful) markers. VAFB SLC-4E has a good success-to-failure ratio. Early CCAFS launches (2010-2014) show more red, while recent launches are predominantly green — confirming improvement over time.



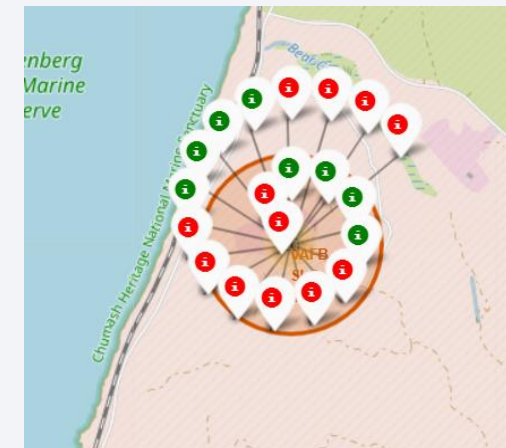
CCAFS LC-40 Success Rate



CCAFS SLC-40 Success Rate



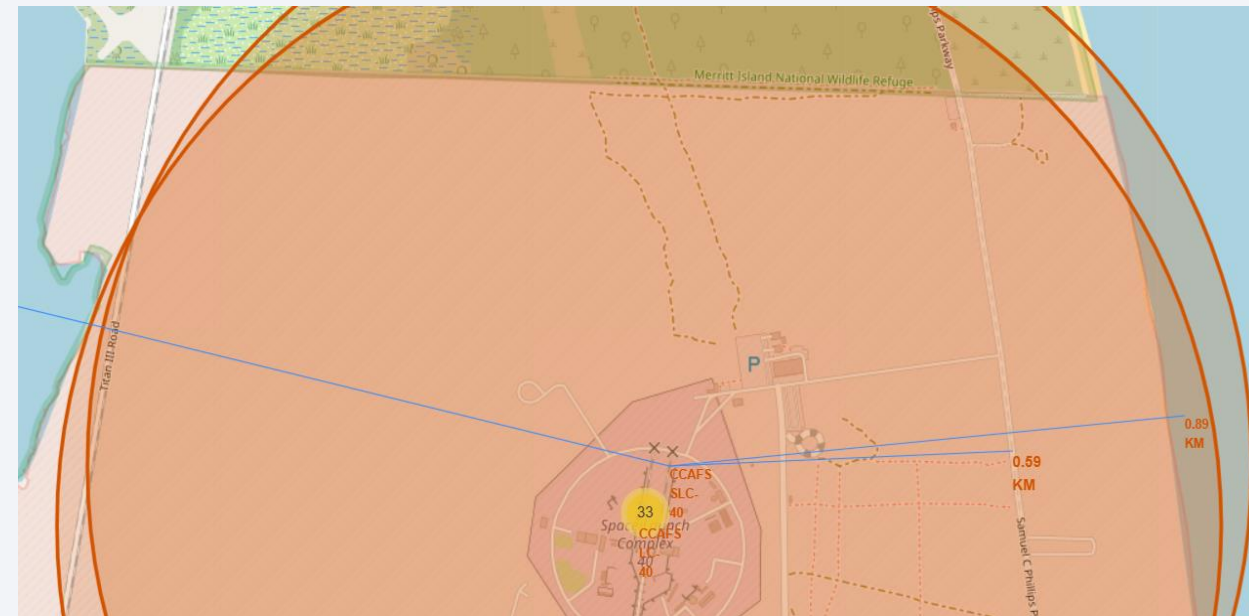
KSC LC-39A Success Rate

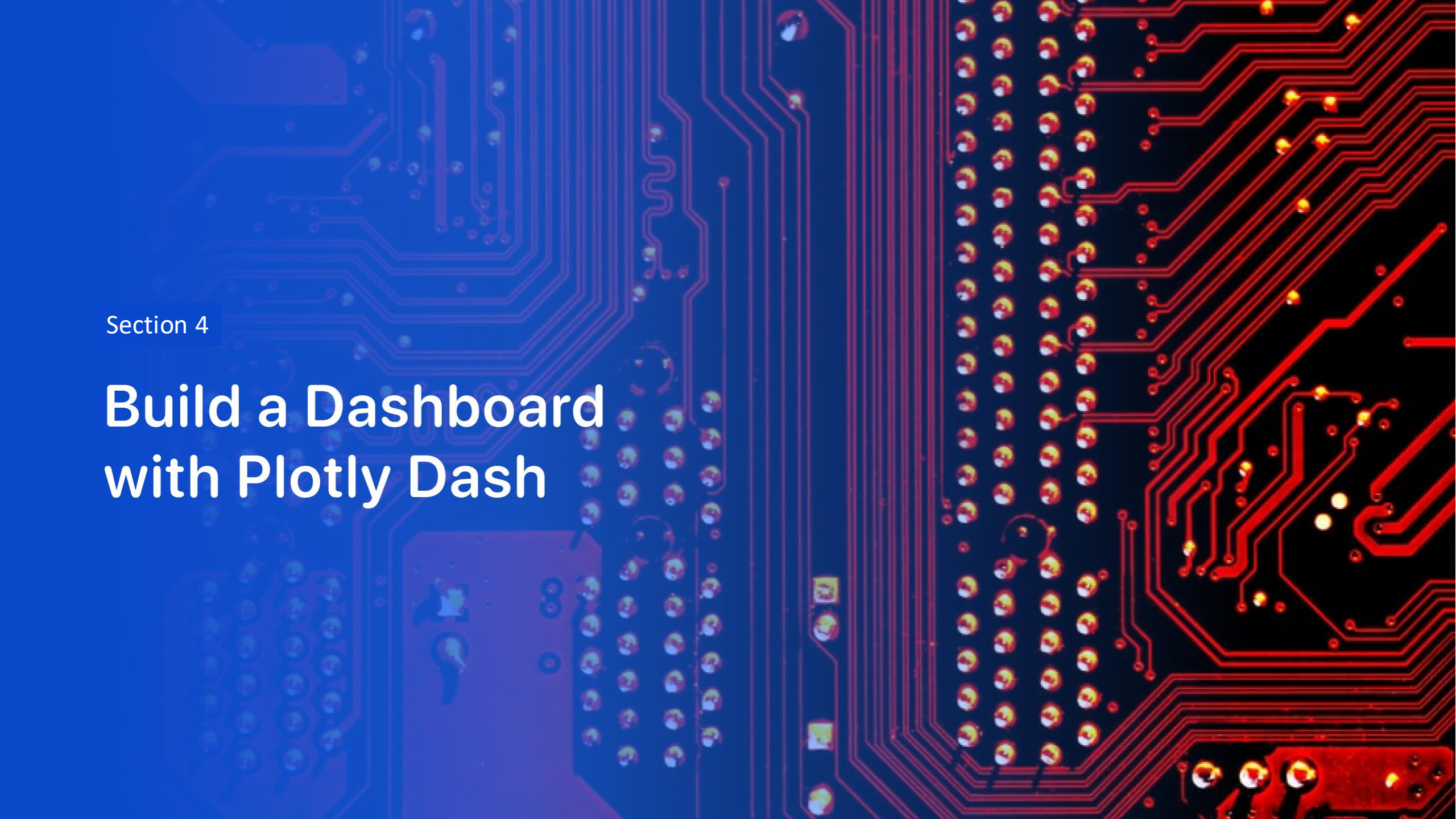


VAFB SLC-4E Success Rate

CCFAS SLC-40 Proximity Analysis

- Distance analysis for CCFAS SLC-40 using PolyLines and distance markers. Measured distances to nearest coastline (Atlantic Ocean), nearby highway, and City of Melbourne. Distances displayed as labeled PolyLines on the map.
- **KEY FINDING:** CCFAS SLC-40 is found approximately 0.89 km from the coastline, ~0.59 km from a highway, and ~23.5 km from the nearest city. This proximity to coast is strategic — failed rockets fall into the ocean. Close to infrastructure for coordination but isolated enough for safety.





Section 4

Build a Dashboard with Plotly Dash



Total Successful Launches

- Dashboard pie chart (dropdown: ALL SITES) shows proportional success count per site. KSC LC-39A leads in success contribution despite fewer total launches.
- KEY FINDING: KSC LC-39A has the highest proportion of successes (~41.7% of all successes). CCAFS SLC-40 has the most launches overall but spread across more failure years.

KSC LC-39A: Highest Success Ratio

- **Payload Mass vs Success (Scatter Plot)**

- Light payloads (2000–3500 kg):**

- > Almost all **class = 1 (success)** — very reliable for lighter missions
 - > Booster versions FT and B4 dominate here

- Medium payloads (3500–5000 kg):**

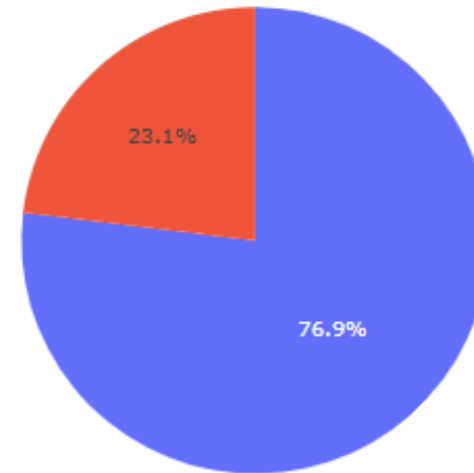
- > Still mostly successful
 - > The Y-axis shows class 0 = failure, class 1 = success
 - > Most failures cluster in the **5000–7000 kg range**

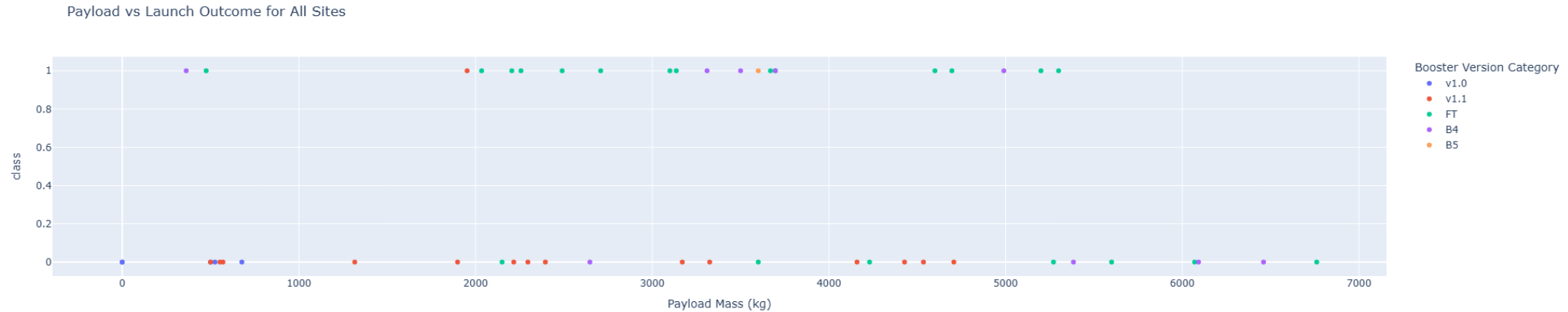
- Heavy payloads (5000+ kg):**

- > A few **class = 0 failures** appear here — heavier payloads are harder to land successfully
 - > Only FT boosters tried the heaviest missions

- **KEY FINDING:** KSC LC-39A has the best success rate likely because it handles a **selective, well-practiced payload range (2000–5000 kg)** with proven booster versions, unlike other sites that try more extreme payload masses.

Success Vs Failure for KSC LC-39A





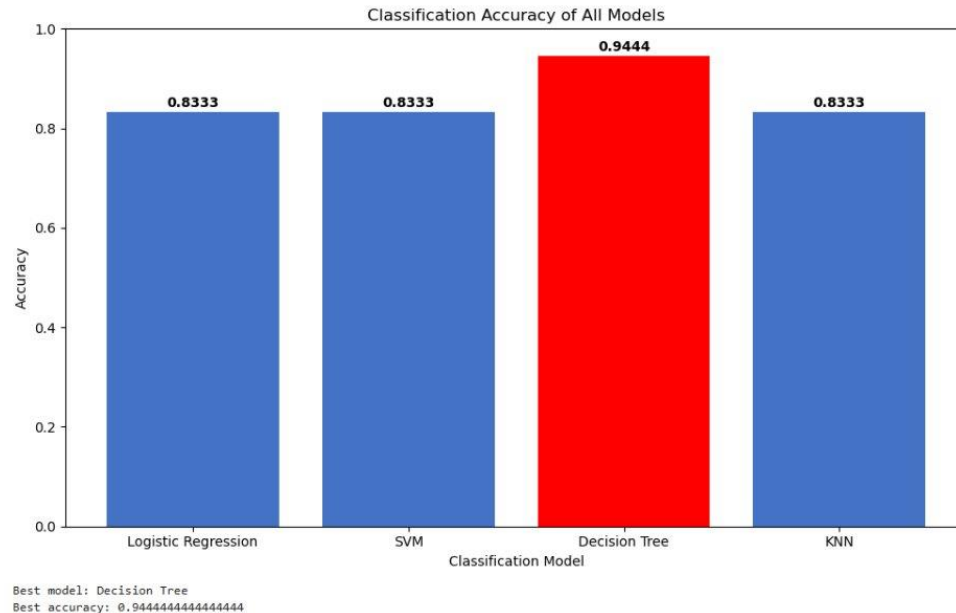
Payload vs. Launch Outcome – All Sites (Range Slider Analysis)

- **Light payloads (0–2,000 kg):**
 - Very few launches, dominated by v1.0 (blue) and v1.1 (red) — older boosters.
 - High failure rate here, reflecting early SpaceX program immaturity.
- **Medium payloads (2,000–5,500 kg):**
 - Highest concentration of launches overall.
 - FT (green), B4 (purple), B5 (orange) dominate.
 - FT booster is the most often used and most successful.
- **Heavy payloads (5,500–10,000 kg):**
 - Even newer boosters struggle at extreme payload masses.

Section 5

Predictive Analysis (Classification)

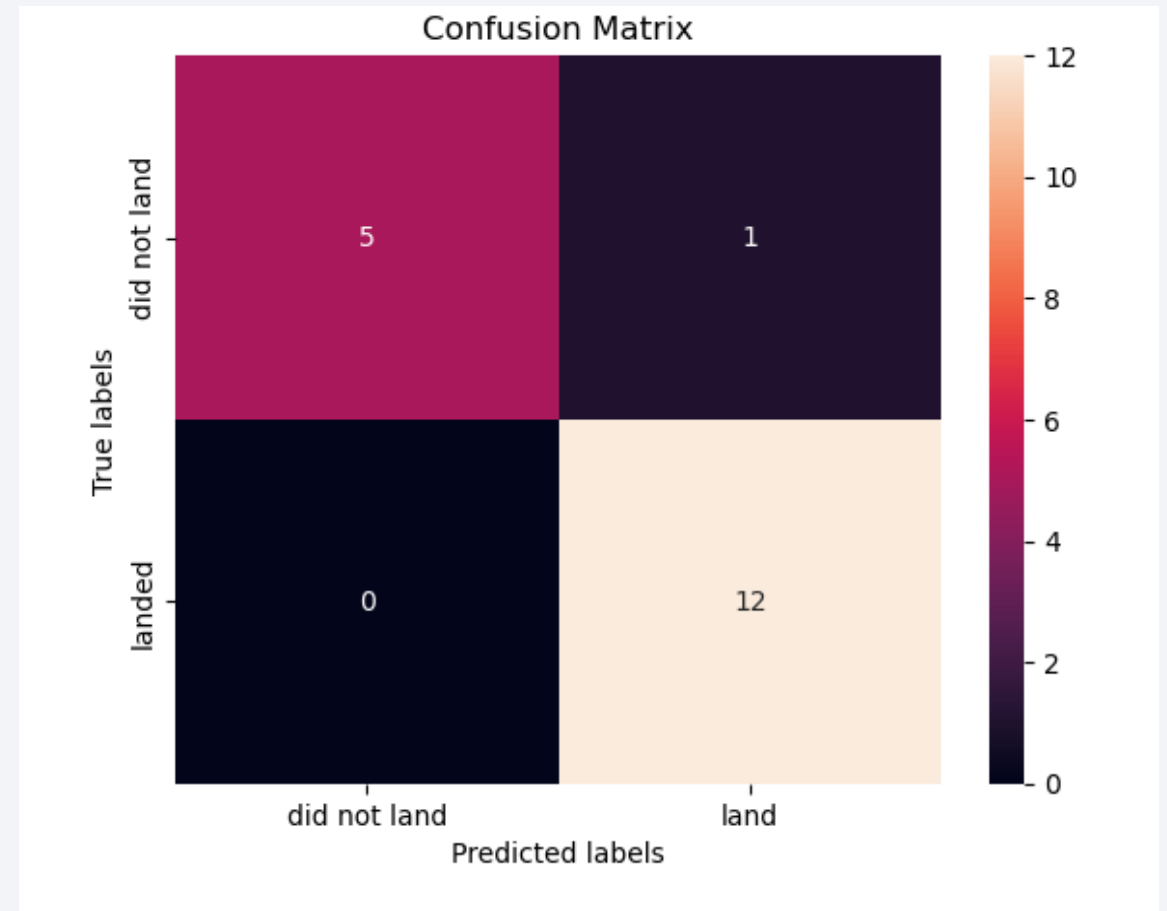
Classification Accuracy



- All four models (Logistic Regression, SVM, Decision Tree, KNN) were tuned using GridSearchCV with cv=10. The Decision Tree classifier achieved the highest test accuracy of **~94.4%**, outperforming all other models. Note: the bar chart from your notebook output should be inserted here as a screenshot.
- **BEST MODEL:** Decision Tree Classifier with 94.4% test accuracy. Optimal hyperparameters: criterion=gini, max_depth=4, max_features=sqrt, min_samples_leaf=2, min_samples_split=2, splitter=best.

Confusion Matrix

- The Decision Tree model achieved **94.4% accuracy** (17/18 correct). The only error was **1 false positive** — one failed landing incorrectly predicted as successful. Critically, the model has **0 false negatives**, meaning it correctly identified every actual successful landing. This makes it highly reliable for SpaceX first stage landing prediction.



Conclusions

- SpaceX Falcon 9 landing success rate improved dramatically from ~0% in 2010 to over 90% by 2020, demonstrating successful mastery of rocket reusability technology.
- KSC LC-39A is the highest-performing launch site with ~76.9% landing success rate. Its infrastructure, derived from Apollo-era facilities, contributes to consistent performance.
- Decision Tree Classifier (max_depth=2, entropy) achieved the best test accuracy of 94.4%, outperforming Logistic Regression, SVM, and KNN (all ~83.3%). Shallow decision trees prevent overfitting while capturing key patterns.
- Payload mass range of 2,000-5,500 kg and B5/FT booster versions are associated with the highest landing success rates. Very heavy payloads (>8,000 kg) are exclusively handled by Block 5 boosters.
- Competitors can use this predictive model to estimate whether SpaceX will reuse the first stage (and thus offer lower prices), informing competitive bidding strategies for commercial launch contracts.

Appendix

- Key assets: (1) SpaceX API collection notebook — JSON normalization, PayloadMass imputation. (2) Web scraping notebook — BeautifulSoup Wikipedia parser. (3) Data wrangling — binary Class label creation from 8 outcome types. (4) EDA visualizations — 6 seaborn charts. (5) SQL queries — 10 analytical queries on SPACEXTBL. (6) Folium interactive map — 4 launch sites with proximity analysis. (7) Plotly Dash app — dropdown, pie chart, range slider, scatter plot. (8) ML notebook — GridSearchCV on 4 classifiers, confusion matrix, Decision Tree winner.

Thank you!

